

■ Search Engine Optimization with AI – Detailed Curriculum (20 Chapters / 4 Units)

◆ Unit 1: Fundamentals of SEO & Artificial Intelligence

Ch	Chapter Title	Detailed Topics	Learning Outcome
1	Introduction to SEO	Definition, Importance, Types (White/Black/Grey Hat), SERP structure	Understand SEO basics and purpose
2	Evolution of Search Engines	Early search engines, Google evolution, crawling & indexing	Learn how search engines developed
3	Basics of AI	AI concepts, ML, NLP, Deep Learning basics	Understand AI foundation
4	Role of AI in SEO	RankBrain, BERT, AI in ranking	Learn AI impact on search results
5	Search Engine Algorithms	Google algorithm updates (Panda, Penguin, Hummingbird)	Analyze ranking factors

◆ Unit 2: On-Page SEO & AI Integration

Ch	Chapter Title	Detailed Topics	Learning Outcome
6	Keyword Research with AI	Tools, long-tail keywords, search intent, AI keyword tools	Perform smart keyword research
7	Content Optimization	SEO writing, readability, keyword density, NLP optimization	Create SEO-friendly content
8	Meta Tags & HTML SEO	Title, meta description, header tags, alt tags	Optimize webpage structure
9	AI Content Generation	AI tools for blogs, prompts, plagiarism, content quality	Use AI responsibly
10	Semantic SEO	Topic clusters, entity SEO, user intent	Improve ranking via context



◆ **Unit 3: Off-Page SEO & Technical SEO**

Ch	Chapter Title	Detailed Topics	Learning Outcome
11	Link Building	Types of backlinks, strategies, outreach	Build quality backlinks
12	Backlink Analysis with AI	Tools, spam detection, competitor analysis	Evaluate backlinks
13	Technical SEO	Crawling, indexing, robots.txt, sitemap.xml	Fix technical issues
14	Website Speed Optimization	Page speed, Core Web Vitals, caching, CDN	Improve performance
15	Mobile SEO	Responsive design, mobile-first indexing	Optimize mobile UX

◆ **Unit 4: Advanced SEO, AI Tools & Analytics**

Ch	Chapter Title	Detailed Topics	Learning Outcome
16	SEO Analytics	Google Analytics, GSC, KPIs, traffic analysis	Measure SEO success
17	AI SEO Tools	Tools like ChatGPT, Surfer SEO, Jasper	Apply AI tools effectively
18	Voice Search SEO	Voice queries, conversational keywords	Optimize for voice search
19	SEO Automation	AI automation tools, bots, workflows	Save time with automation
20	Future of SEO & AI	AI trends, SGE, personalization, ethical SEO	Prepare for future



Course Outcomes (CO1 – CO12)



◆ Unit 1 – Course Outcomes

CO No.	Course Outcome Statement
CO1	Understand the fundamental concepts of SEO, search engines, and their working mechanisms
CO2	Analyze the role of Artificial Intelligence in modern search engine optimization
CO3	Evaluate different search engine algorithms and their impact on website ranking

◆ Unit 2 – Course Outcomes

CO No.	Course Outcome Statement
CO4	Apply keyword research techniques using AI tools for effective SEO strategies
CO5	Develop optimized web content using on-page SEO techniques and AI-based tools
CO6	Implement semantic SEO and user intent strategies for better search visibility

◆ Unit 3 – Course Outcomes

CO No.	Course Outcome Statement
CO7	Demonstrate knowledge of link building and off-page SEO techniques
CO8	Perform technical SEO tasks including crawling, indexing, and website optimization
CO9	Analyze website performance using Core Web Vitals and mobile SEO techniques

◆ Unit 4 – Course Outcomes

CO No.	Course Outcome Statement
CO10	Utilize SEO analytics tools to measure and improve website performance
CO11	Apply AI-based SEO tools and automation techniques in real-world scenarios
CO12	Assess future trends and advancements in SEO and AI for strategic decision-making

Practical + Lab Work (Very Important for IT Students)

Add this to your curriculum (makes it industry-ready):

Practical No.	Task
1	Keyword research using AI tools
2	On-page SEO optimization of a webpage
3	Create SEO-friendly blog using AI
4	Backlink analysis of competitor site
5	Technical SEO audit (using tools)
6	Website speed testing
7	Google Search Console analysis
8	SEO report creation

Assessment Pattern (MDU Style)

Type	Marks	Description
Internal	30	Assignments + Practical
External	70	Written Exam
Section A	10	1-mark questions
Section B	20	Short Questions
Section C	40	Long Questions



Detail Chapter Wise Topic and Course Outcome

◆ Unit 1 – Chapter 1 (Introduction to SEO and AI)

S. No.	Major Topic	Sub Topics	Sub Course Outcome (SCO)
1	Introduction to SEO	Definition, Need, Scope	SCO1.1: Understand the basic concept and importance of SEO
2	Types of SEO	On-page, Off-page, Technical SEO	SCO1.2: Differentiate between types of SEO
3	Search Engine Basics	Definition, Examples, Working overview	SCO1.3: Explain how search engines function
4	SERP (Search Engine Result Page)	Organic vs Paid results, Features	SCO1.4: Identify components of SERP
5	Introduction to AI	Definition, History, Applications	SCO1.5: Understand AI fundamentals
6	AI Techniques	Machine Learning, NLP basics	SCO1.6: Describe basic AI techniques used in SEO
7	Role of AI in SEO	Ranking, personalization, automation	SCO1.7: Analyze how AI enhances SEO processes
8	SEO vs AI-driven SEO	Traditional vs Modern SEO	SCO1.8: Compare traditional SEO with AI-based SEO
9	Importance of Keywords	Keyword concept, types	SCO1.9: Explain role of keywords in SEO
10	User Intent	Informational, Navigational, Transactional	SCO1.10: Interpret user search intent
11	Crawling & Indexing	Bots, spiders, indexing process	SCO1.11: Describe crawling and indexing process
12	Ranking Factors	Content, backlinks, UX signals	SCO1.12: Identify major ranking factors
13	Ethical SEO Practices	White Hat vs Black Hat SEO	SCO1.13: Apply ethical SEO techniques



◆ **Unit 1 – Chapter 2 (Evolution of Search Engines)**

S. No.	Major Topic	Sub Topics	Sub Course Outcome (SCO)
1	Introduction to Search Engines	Definition, Purpose, Examples	SCO2.1: Understand the concept and purpose of search engines
2	Early Search Engines	Archie, AltaVista, Yahoo Directory	SCO2.2: Describe early search engines and their limitations
3	Limitations of Early Systems	Keyword stuffing, poor relevance	SCO2.3: Analyze drawbacks of early search engines
4	Emergence of Google	Founders, PageRank concept	SCO2.4: Explain the rise of Google and its innovations
5	Evolution of Algorithms	Basic → Advanced algorithms	SCO2.5: Understand algorithm evolution over time
6	Crawling Mechanism	Web crawlers, bots, spiders	SCO2.6: Describe crawling process in search engines
7	Indexing Process	Data storage, indexing techniques	SCO2.7: Explain how web pages are indexed
8	Ranking Mechanism	Ranking signals, relevance factors	SCO2.8: Identify factors affecting search rankings
9	Introduction to PageRank	Link-based ranking system	SCO2.9: Understand PageRank working principle
10	Shift to Semantic Search	Meaning-based search, context	SCO2.10: Analyze transition to semantic search
11	AI Integration in Search	Machine learning, NLP, RankBrain	SCO2.11: Evaluate AI's role in search evolution
12	Mobile & Voice Search Evolution	Mobile-first indexing, voice queries	SCO2.12: Understand modern search trends
13	Future Trends	AI search, personalization, SGE	SCO2.13: Predict future of search engines



◆ **Unit 1 – Chapter 3 (Basics of Artificial Intelligence)**

S. No.	Major Topic	Sub Topics	Sub Course Outcome (SCO)
1	Introduction to AI	Definition, meaning, scope	SCO3.1: Understand the basic concept of Artificial Intelligence
2	History of AI	Early development, milestones	SCO3.2: Describe the evolution and history of AI
3	Types of AI	Narrow AI, General AI, Super AI	SCO3.3: Differentiate between types of AI
4	AI vs Human Intelligence	Decision making, learning ability	SCO3.4: Compare AI with human intelligence
5	Machine Learning	Definition, supervised & unsupervised learning	SCO3.5: Explain basics of Machine Learning
6	Deep Learning	Neural networks, layers, applications	SCO3.6: Understand deep learning concepts
7	Natural Language Processing (NLP)	Text processing, language understanding	SCO3.7: Describe role of NLP in AI systems
8	AI Algorithms	Basic algorithms, logic-based systems	SCO3.8: Identify common AI algorithms
9	Applications of AI	Healthcare, education, SEO, automation	SCO3.9: Analyze real-world applications of AI
10	AI in Digital Marketing	Personalization, chatbots, recommendations	SCO3.10: Evaluate use of AI in marketing & SEO
11	Advantages of AI	Efficiency, automation, accuracy	SCO3.11: Identify benefits of AI systems
12	Limitations of AI	Cost, bias, lack of creativity	SCO3.12: Analyze challenges and limitations of AI



◆ **Unit 1 – Chapter 4 (Role of AI in SEO)**

S. No.	Major Topic	Sub Topics	Sub Course Outcome (SCO)
1	Introduction to AI in SEO	Definition, importance, overview	SCO4.1: Understand the role of AI in modern SEO
2	AI in Search Algorithms	RankBrain, BERT, AI ranking signals	SCO4.2: Analyze how AI influences search algorithms
3	Keyword Research using AI	AI tools, keyword suggestions, trends	SCO4.3: Apply AI techniques for keyword research
4	AI-based Content Optimization	NLP content, readability, keyword placement	SCO4.4: Develop optimized content using AI
5	Search Intent Analysis	User behavior, intent prediction	SCO4.5: Interpret user intent using AI tools
6	AI in Technical SEO	Site audit, crawling issues detection	SCO4.6: Utilize AI for technical SEO improvements
7	Personalization in SEO	User data, behavior-based results	SCO4.7: Explain AI-driven personalization in search
8	Voice Search Optimization	Conversational queries, AI assistants	SCO4.8: Optimize content for voice-based search
9	AI-powered Chatbots	Customer interaction, engagement	SCO4.9: Evaluate chatbot role in SEO & UX
10	Predictive SEO	Trend forecasting, data-driven decisions	SCO4.10: Analyze predictive capabilities of AI in SEO
11	Automation in SEO	Automated reports, content, audits	SCO4.11: Implement automation using AI tools
12	Ethical Issues in AI SEO	Bias, data privacy, misuse of AI	SCO4.12: Identify ethical concerns in AI-driven SEO



◆ **Unit 1 – Chapter 5 (Search Engine Algorithms)**

S. No.	Major Topic	Sub Topics	Sub Course Outcome (SCO)
1	Introduction to Algorithms	Definition, role in SEO, importance	SCO5.1: Understand the concept of search engine algorithms
2	Working of Algorithms	Input, processing, ranking output	SCO5.2: Explain how algorithms determine search rankings
3	Major Algorithm Updates	Panda, Penguin, Hummingbird overview	SCO5.3: Identify major Google algorithm updates
4	Google Panda Algorithm	Content quality, duplicate content	SCO5.4: Analyze impact of Panda on content quality
5	Google Penguin Algorithm	Backlinks, spam links	SCO5.5: Evaluate link-based ranking penalties
6	Google Hummingbird	Semantic search, intent-based results	SCO5.6: Understand semantic search improvements
7	RankBrain Algorithm	Machine learning in ranking	SCO5.7: Explain AI-based ranking via RankBrain
8	BERT Algorithm	Natural language understanding	SCO5.8: Analyze impact of BERT on search queries
9	Core Algorithm Updates	Broad core updates, ranking changes	SCO5.9: Interpret effects of core updates
10	Ranking Factors	Content, backlinks, UX signals	SCO5.10: Identify key ranking factors
11	Penalties & Recovery	Manual penalties, recovery strategies	SCO5.11: Apply methods to recover from penalties
12	Future Algorithm Trends	AI-based algorithms, automation	SCO5.12: Predict future developments in algorithms



◆ **Unit 2 – Chapter 6 (Keyword Research with AI)**

S. No.	Major Topic	Sub Topics	Sub Course Outcome (SCO)
1	Introduction to Keywords	Definition, importance in SEO	SCO6.1: Understand the concept and role of keywords in SEO
2	Types of Keywords	Short-tail, Long-tail, LSI keywords	SCO6.2: Differentiate between various types of keywords
3	Keyword Research Process	Steps, strategy, planning	SCO6.3: Apply systematic keyword research process
4	Search Intent Analysis	Informational, Navigational, Transactional	SCO6.4: Interpret user intent behind search queries
5	AI in Keyword Research	AI tools, automation, data analysis	SCO6.5: Analyze how AI improves keyword research
6	Keyword Research Tools	Google Keyword Planner, AI tools	SCO6.6: Use tools for effective keyword research
7	Competitor Keyword Analysis	Competitor strategy, gap analysis	SCO6.7: Evaluate competitor keywords using AI
8	Keyword Difficulty & Volume	Search volume, competition level	SCO6.8: Assess keyword difficulty and ranking potential
9	Long-tail Keywords with AI	AI suggestions, niche targeting	SCO6.9: Generate long-tail keywords using AI tools
10	Trend Analysis	Seasonal trends, Google Trends, AI predictions	SCO6.10: Analyze keyword trends using AI
11	Keyword Clustering	Grouping keywords, topic clusters	SCO6.11: Organize keywords for content optimization
12	Keyword Placement Strategy	Title, headings, content, meta tags	SCO6.12: Implement proper keyword placement techniques



◆ **Unit 2 – Chapter 7 (Content Optimization)**

S. No.	Major Topic	Sub Topics	Sub Course Outcome (SCO)
1	Introduction to Content Optimization	Definition, importance in SEO	SCO7.1: Understand the concept of content optimization
2	SEO Content Writing	Writing for users + search engines	SCO7.2: Develop SEO-friendly content writing skills
3	Keyword Optimization	Keyword placement, density, stuffing avoidance	SCO7.3: Apply proper keyword optimization techniques
4	Content Structure	Headings (H1–H6), paragraphs, formatting	SCO7.4: Organize content for better readability
5	Readability & UX	Simple language, readability score, engagement	SCO7.5: Improve user experience through readable content
6	AI-based Content Optimization	NLP tools, AI suggestions, automation	SCO7.6: Utilize AI tools for optimizing content
7	Content Quality Factors	Originality, relevance, depth, freshness	SCO7.7: Evaluate content quality for better ranking
8	Multimedia Optimization	Images, videos, alt tags	SCO7.8: Optimize multimedia elements in content
9	Internal Linking	Linking strategy within website	SCO7.9: Implement internal linking for SEO benefits
10	External Linking	Outbound links, authority sources	SCO7.10: Apply external linking techniques effectively
11	Content Update Strategy	Refreshing old content, updating keywords	SCO7.11: Maintain content relevance through updates
12	Content Performance Analysis	Metrics, engagement, AI analytics tools	SCO7.12: Analyze content performance using SEO tools



◆ **Unit 2 – Chapter 8 (Meta Tags & HTML SEO)**

S. No.	Major Topic	Sub Topics	Sub Course Outcome (SCO)
1	Introduction to HTML SEO	Role of HTML in SEO, basics	SCO8.1: Understand the importance of HTML in SEO
2	Meta Tags Overview	Definition, purpose, types	SCO8.2: Identify different meta tags used in SEO
3	Title Tag Optimization	Length, keywords, best practices	SCO8.3: Create effective and optimized title tags
4	Meta Description	Writing, CTR improvement, guidelines	SCO8.4: Develop optimized meta descriptions
5	Meta Keywords (Deprecated)	Concept, limitations, history	SCO8.5: Understand the relevance of meta keywords
6	Heading Tags (H1–H6)	Structure, hierarchy, usage	SCO8.6: Apply proper heading tag structure
7	Image Optimization	Alt text, file names, compression	SCO8.7: Optimize images for SEO
8	URL Structure	SEO-friendly URLs, slugs	SCO8.8: Design optimized URL structures
9	Robots Meta Tag	Index, noindex, follow, nofollow	SCO8.9: Control crawling and indexing using robots tag
10	Canonical Tag	Duplicate content handling	SCO8.10: Apply canonical tags to avoid duplication issues
11	Open Graph Tags	Social media optimization	SCO8.11: Implement OG tags for better sharing
12	Schema Markup	Structured data, rich snippets	SCO8.12: Use schema markup to enhance search visibility



◆ **Unit 2 – Chapter 9 (AI-based Content Generation)**

S. No.	Major Topic	Sub Topics	Sub Course Outcome (SCO)
1	Introduction to AI Content	Definition, importance in SEO	SCO9.1: Understand the concept of AI-based content generation
2	AI Content Generation Tools	ChatGPT, Jasper, Copy.ai	SCO9.2: Identify and use AI tools for content creation
3	Working of AI Models	NLP, training data, text prediction	SCO9.3: Explain how AI generates content
4	Prompt Engineering	Writing effective prompts, instructions	SCO9.4: Apply prompt techniques for better output
5	Types of AI Content	Blogs, articles, product descriptions	SCO9.5: Generate different types of SEO content using AI
6	SEO Optimization with AI	Keyword integration, readability	—
7	Content Quality Control	Editing, fact-checking, plagiarism	SCO9.6: Evaluate and improve AI-generated content quality
8	Advantages of AI Content	Speed, scalability, automation	—
9	Limitations of AI Content	Accuracy issues, lack of originality	SCO9.7: Analyze limitations of AI-generated content
10	Ethical Considerations	Bias, copyright, misuse of AI	SCO9.8: Identify ethical issues in AI content usage



◆ **Unit 2 – Chapter 10 (User Intent & Semantic SEO)**

S. No.	Major Topic	Sub Topics	Sub Course Outcome (SCO)
1	Introduction to User Intent	Definition, importance in SEO	SCO10.1: Understand the concept of user intent
2	Types of User Intent	Informational, Navigational, Transactional, Commercial	SCO10.2: Differentiate between types of user intent
3	Intent-based Keyword Research	Mapping keywords to intent	SCO10.3: Apply intent-based keyword strategies
4	Introduction to Semantic SEO	Meaning-based search, context	SCO10.4: Understand semantic SEO principles
5	Evolution to Semantic Search	Keyword → Context-based search	—
6	Entity-based SEO	Entities, relationships, knowledge graph	SCO10.5: Explain entity-based optimization
7	Topic Clusters	Pillar content, cluster strategy	SCO10.6: Organize content using topic clusters
8	Latent Semantic Indexing (LSI)	Related keywords, context building	—
9	AI in Semantic SEO	NLP, search intent prediction	SCO10.7: Analyze role of AI in semantic search
10	Content Optimization for Intent	Matching content with user needs	SCO10.8: Optimize content based on user intent
11	Voice & Conversational Search	Natural queries, question-based search	—
12	Practical Implementation	Real examples, SEO strategy building	SCO10.9: Apply semantic SEO strategies in practice



Unit – 3 | Chapter – 11: Link Building Strategies

S. No.	Major Topic	Sub Topics	Sub Course Outcome (SCO)
1	Introduction to Link Building	Definition, role in SEO	SCO11.1: Basics of link building
2	Importance of Backlinks	Ranking factor, authority	SCO11.2: Role of backlinks in SEO
3	Types of Backlinks	DoFollow, NoFollow	SCO11.3: Types of backlinks
4	Quality vs Quantity	High vs low quality links	SCO11.4: Link quality evaluation
5	White Hat Techniques	Ethical methods	SCO11.5: White hat strategies
6	Black Hat Techniques	Spam links, link farms	SCO11.6: Black hat risks
7	Guest Posting	Content contribution	SCO11.7: Guest posting method
8	Directory & Bookmarking	Submission sites	SCO11.8: Off-page submission methods
9	Broken Link Building	Finding & fixing links	SCO11.9: Broken link strategy
10	Anchor Text Optimization	Types, keyword usage	SCO11.10: Anchor text usage
11	Link Building Tools	Basic + AI tools	SCO11.11: Tools for link building
12	Common Mistakes	Over-optimization, spam	SCO11.12: Link building errors



Unit – 3 | Chapter – 12: Backlink Analysis with AI

S. No.	Major Topic	Sub Topics	Sub Course Outcome (SCO)
1	Introduction to Backlink Analysis	Definition, importance in SEO	SCO12.1: Basics of backlink analysis
2	Backlink Metrics	Domain Authority, Page Authority, Spam Score	SCO12.2: Key backlink metrics
3	Types of Backlinks Analysis	DoFollow, NoFollow, toxic links	SCO12.3: Types of backlinks evaluation
4	AI in Backlink Analysis	AI tools, automation, data insights	SCO12.4: AI role in backlink analysis
5	Competitor Backlink Analysis	Identifying competitor strategies	SCO12.5: Competitor backlink study
6	Toxic Backlink Detection	Spam links, harmful backlinks	SCO12.6: Detection of toxic backlinks
7	Backlink Audit Process	Steps, tools, reporting	SCO12.7: Backlink audit method
8	Link Disavow Process	Removing harmful links	SCO12.8: Disavow technique
9	Backlink Monitoring	Tracking links over time	SCO12.9: Backlink monitoring methods

Additional Topics →

10. Use of ANH tools like

- [Global Github Pages - Advance SEO Inspection Tool](#)
- [SEO Notes and URL Inspection tool for ANH Pages](#)
- [Links Inspection Tools for All Github Pages](#)



◆ Unit 3 – Chapter 13 (Technical SEO)

S. No.	Major Topic	Sub Topics	Sub Course Outcome (SCO)
1	Introduction to Technical SEO	Definition, role in SEO	SCO13.1: Basics of technical SEO
2	Website Crawling	Crawlers, bots, crawl process	SCO13.2: Crawling mechanism
3	Indexing	Indexing process, search engine database	SCO13.3: Indexing concepts
4	Robots.txt	Syntax, blocking pages	SCO13.4: Robots.txt usage
5	XML Sitemap	Structure, submission, importance	SCO13.5: Sitemap implementation
6	URL Structure	Clean URLs, SEO-friendly design	SCO13.6: URL optimization
7	Canonicalization	Duplicate content handling	SCO13.7: Canonical tag usage
8	Website Architecture	Internal linking, hierarchy	SCO13.8: Site structure optimization
9	HTTPS & Security	SSL, secure websites	SCO13.9: Website security basics
10	Technical SEO Tools	Audit tools, AI-based tools overview	SCO13.10: Technical SEO tools usage

➔ Hands on with Practical Works ➔

- Use of Meta Tags and Implementing them on our Portfolio.
- Introducing Breadcrumbs , Twitter Cards , OG , Cannonical logo / url , Use of <link rel= “”> tags .
- Creating & Submitting Sitemap on GSC.
- Complete Use of GSC.



◆ **Unit 3 – Chapter 14 (Website Speed Optimization)**

S. No.	Major Topic	Sub Topics	Sub Course Outcome (SCO)
1	Introduction to Website Speed	Definition, importance in SEO	SCO14.1: Basics of website speed optimization
2	Page Load Time Factors	Server, images, scripts	SCO14.2: Factors affecting page speed
3	Core Web Vitals	LCP, FID, CLS metrics	SCO14.3: Core Web Vitals understanding
4	Image Optimization	Compression, formats (WebP, JPEG)	SCO14.4: Image optimization techniques
5	Code Optimization	Minification of CSS, JS, HTML	SCO14.5: Code optimization methods
6	Caching Techniques	Browser cache, server cache	SCO14.6: Caching implementation
7	Content Delivery Network (CDN)	Distributed servers, faster delivery	SCO14.7: CDN usage
8	Mobile Speed Optimization	Mobile-first performance	SCO14.8: Mobile speed improvement
9	Speed Testing Tools	PageSpeed Insights, GTmetrix	SCO14.9: Speed testing tools usage
10	Performance Best Practices	Lazy loading, script optimization	SCO14.10: Website performance practices



◆ **Unit 3 – Chapter 15 (Mobile SEO & Core Web Vitals)**

S. No.	Major Topic	Sub Topics	Sub Course Outcome (SCO)
1	Introduction to Mobile SEO	Definition, importance	SCO15.1: Basics of mobile SEO
2	Mobile-First Indexing	Concept, Google approach	SCO15.2: Mobile-first indexing
3	Responsive Web Design	Flexible layout, media queries	SCO15.3: Responsive design principles
4	Mobile Usability	UX, readability, navigation	SCO15.4: Mobile usability factors
5	Page Speed for Mobile	Load time, optimization	SCO15.5: Mobile speed optimization
6	Core Web Vitals Overview	LCP, FID, CLS introduction	SCO15.6: Core Web Vitals basics
7	Largest Contentful Paint (LCP)	Loading performance metric	SCO15.7: LCP understanding
8	First Input Delay (FID)	Interactivity measurement	SCO15.8: FID concept
9	Cumulative Layout Shift (CLS)	Visual stability metric	SCO15.9: CLS concept
10	Mobile SEO Tools	Testing tools, performance analysis	SCO15.10: Mobile SEO tools usage

➔ Hands on with Practical ➔

- Adding Mobile first UI and Responsiveness to your Portfolio.
- Use @media in CSS properties to apply styles across multi media with first focus on Mobile UI.
- Check your website UI and Responsive across various device media using Inspection tool in Chrome , Microsoft Edge or other browser .



◆ Unit 4 – Chapter 16 (SEO Analytics & Metrics)

S. No.	Major Topic	Sub Topics	Sub Course Outcome (SCO)
1	Introduction to SEO Analytics	Definition, importance	SCO16.1: Basics of SEO analytics
2	Key SEO Metrics	Traffic, impressions, CTR	SCO16.2: SEO performance metrics
3	Organic Traffic Analysis	Sources, behavior, trends	SCO16.3: Organic traffic analysis
4	Keyword Performance Tracking	Ranking, keyword position	SCO16.4: Keyword tracking methods
5	Bounce Rate & Engagement	User interaction, session time	SCO16.5: User engagement metrics
6	Conversion Rate	Leads, goals, ROI	SCO16.6: Conversion analysis
7	Google Analytics Overview	Reports, dashboards	SCO16.7: Analytics tools usage
8	Google Search Console	Indexing, performance reports	SCO16.8: Search Console analysis
9	AI in SEO Analytics	Predictive analysis, automation	SCO16.9: AI role in analytics
10	Reporting & Insights	SEO reports, data interpretation	SCO16.10: SEO reporting skills

- ➔ Monthly Analysis of our Sites on GSC after Submitting to track Metrics ➔
- Learn about Engagement , Impression , Click Rate , AI Overview , etc.
 - Building your identity and make sure Google Bots consider you as a entity.
 - Verify your Sites with multi verification methods and give ownership access to authorized person only.

◆ Unit 4 – Chapter 17 (AI SEO Tools)

S. No.	Major Topic	Sub Topics	Sub Course Outcome (SCO)
1	Introduction to AI SEO Tools	Definition, importance	SCO17.1: Basics of AI SEO tools
2	Content Generation Tools	AI writing, blog creation	SCO17.2: AI content tools usage
3	Keyword Research Tools	AI keyword suggestions, trends	SCO17.3: AI keyword tools
4	On-Page Optimization Tools	Content scoring, recommendations	SCO17.4: On-page AI tools
5	Technical SEO Tools	Site audit, error detection	SCO17.5: Technical SEO tools
6	Backlink Analysis Tools	Link analysis, monitoring	SCO17.6: Backlink tools usage
7	Competitor Analysis Tools	Strategy comparison, insights	SCO17.7: Competitor analysis tools
8	AI Chatbots & Assistants	Automation, user interaction	SCO17.8: AI chatbot usage
9	Automation Tools	Workflow automation, reporting	SCO17.9: SEO automation tools
10	Advantages & Limitations	Benefits, risks, accuracy	SCO17.10: Evaluation of AI tools

➔ Learn about More Free / Paid Tools ➔

- Use many tools to track and monitor your sites .
- Use of AI assistant and Prompt Engineering to build Interactive Dashboards .
- Use AI codes to understand logic and don't copy – paste without understanding it .
- Use Cases of Backlinks and Why to avoid unnecessary Backlinks ?



◆ **Unit 4 – Chapter 18 (Voice Search Optimization)**

S. No.	Major Topic	Sub Topics	Sub Course Outcome (SCO)
1	Introduction to Voice Search	Definition, growth, importance	SCO18.1: Basics of voice search
2	Working of Voice Search	Speech recognition, NLP	SCO18.2: Voice search mechanism
3	Voice Search vs Text Search	Differences, query style	SCO18.3: Voice vs text comparison
4	Conversational Keywords	Long-tail, natural language queries	SCO18.4: Conversational keyword usage
5	User Intent in Voice Search	Question-based queries	SCO18.5: Intent analysis in voice search
6	Featured Snippets	Position zero, direct answers	SCO18.6: Featured snippet optimization
7	Local SEO for Voice Search	"Near me" queries, local results	SCO18.7: Local voice SEO
8	Mobile Optimization	Mobile-first, fast loading	SCO18.8: Mobile voice optimization
9	AI Assistants	Siri, Alexa, Google Assistant	SCO18.9: Role of AI assistants
10	Voice Search Best Practices	Structured data, FAQs	SCO18.10: Voice SEO strategies

➔ Hands on with Practicals ➔

- Introducing Bootstrap and Its importance in Responsive Web Design .
- Use of Media Queries to implement mobile optimization and UI adaptability according to various device width .
- Implementing AI features in our Sites .



◆ **Unit 4 – Chapter 19 (SEO Automation using AI)**

S. No.	Major Topic	Sub Topics	Sub Course Outcome (SCO)
1	Introduction to SEO Automation	Definition, importance	SCO19.1: Basics of SEO automation
2	Role of AI in Automation	Machine learning, NLP	SCO19.2: AI in SEO automation
3	Automated Keyword Research	AI-based keyword discovery	SCO19.3: Keyword automation
4	Content Automation	AI content generation, optimization	SCO19.4: Content automation tools
5	Technical SEO Automation	Site audits, error fixing	SCO19.5: Technical automation
6	Backlink Automation	Link monitoring, outreach tools	SCO19.6: Backlink automation
7	Rank Tracking Automation	Automated ranking reports	SCO19.7: Rank tracking tools
8	SEO Reporting Automation	Dashboards, automated reports	SCO19.8: Reporting automation
9	Benefits of SEO Automation	Efficiency, scalability	SCO19.9: Advantages of automation
10	Limitations & Risks	Over-reliance, accuracy issues	SCO19.10: Risks of automation

➔ Hands on with Practical Like ➔

- Use an AI tool to generate SEO keywords for a selected topic .
- Create SEO-optimized content using AI and improve it with suggestion .
- Perform an automated SEO audit of a website and identify key issues .
- Track keyword rankings automatically using an SEO tool .
- Generate SEO reports automatically using AI-based dashboards .



◆ **Unit 4 – Chapter 20 (Future of SEO with AI)**

S. No.	Major Topic	Sub Topics	Sub Course Outcome (SCO)
1	Introduction to Future SEO	Evolution of SEO with AI	SCO20.1: Future SEO basics
2	AI-driven Search Engines	Smart algorithms, personalization	SCO20.2: AI search evolution
3	Voice & Visual Search	Voice queries, image search	SCO20.3: Emerging search methods
4	Predictive SEO	User behavior prediction	SCO20.4: Predictive SEO concepts
5	Hyper-Personalization	Customized search results	SCO20.5: Personalized SEO strategies
6	AI Content Evolution	Auto-generated, adaptive content	SCO20.6: Future content trends
7	Automation in SEO	Fully automated workflows	SCO20.7: SEO automation future
8	Ethical & Privacy Issues	Data privacy, AI ethics	SCO20.8: Ethical considerations
9	Role of Human SEO Experts	Strategy, creativity, control	SCO20.9: Human vs AI role
10	Future SEO Best Practices	Adaptation, continuous learning	SCO20.10: Future SEO strategies

➔ Hands on with Practical ➔

- Analyze how AI-based search results differ from traditional search results .
- Perform a voice search query and study the type of results displayed .
- Use an AI tool to generate personalized SEO content for a target audience .
- Identify ethical and privacy issues in AI-based SEO using real examples .



SEO with AI – Practical File (12 Practicals)

S. No.	Major Topic	Sub Topics	Sub Course Outcome (SCO)
1	SEO Basics & Website Analysis	SEO audit, website structure, indexing	SCO-P1: Understand complete SEO workflow
2	Keyword Research (AI-Based)	Keyword tools, search intent, trends	SCO-P2: Perform AI-based keyword research
3	On-Page SEO Optimization	Meta tags, headings, content SEO	SCO-P3: Apply on-page optimization
4	Technical SEO Audit	Sitemap, robots.txt, URL structure	SCO-P4: Analyze technical SEO factors
5	Content Optimization using AI	AI writing, readability, keyword usage	SCO-P5: Optimize content using AI
6	Backlink Building & Analysis	Link building, backlink audit, tools	SCO-P6: Manage backlinks effectively
7	Competitor SEO Analysis	Keyword gap, backlink comparison	SCO-P7: Perform competitor analysis
8	Website Speed Optimization	Core Web Vitals, caching, CDN	SCO-P8: Improve website performance
9	Mobile SEO & User Experience	Mobile-first design, usability	SCO-P9: Optimize for mobile SEO
10	SEO Analytics & Reporting	Traffic, CTR, Google Analytics	SCO-P10: Analyze SEO performance
11	SEO Automation using AI	Automation tools, reporting, tracking	SCO-P11: Implement SEO automation
12	Future SEO & Voice Search	Voice SEO, AI trends, personalization	SCO-P12: Understand future SEO trends